

✓ EXPONENTS & SURDS

Rational exponent

$a^{\frac{m}{n}} = \sqrt[n]{a^m}$

Surd laws

$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$

$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$

Rationalising

$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$

with a binomial

$\frac{1}{\sqrt{a} + \sqrt{b}} \times \frac{\sqrt{a} - \sqrt{b}}{\sqrt{a} - \sqrt{b}}$

Multiply by the conjugate, not by the same expression — that is what clears the surd, because $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$.

Σ QUADRATIC NUMBER PATTERNS

General term

$T_n = an^2 + bn + c$

Second difference

$2a$

First difference

$3a + b$

First term

$a + b + c$

Build the three equations from the top of the pattern and solve in order — a from the second difference, then b, then c. Guessing T_n and checking is slower and loses method marks.

Δ EQUATIONS & NATURE OF ROOTS

Quadratic formula

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminant

$\Delta = b^2 - 4ac$

$\Delta > 0$

two unequal real roots

$\Delta = 0$

two equal real roots

$\Delta < 0$

non-real roots

Δ a perfect square

roots are rational

"Real and unequal" and "rational" are different questions. $\Delta > 0$ answers the first; whether Δ is a perfect square answers the second, and the exam asks for both.

⊙ ANALYTICAL GEOMETRY

Distance

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint

$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Gradient

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Parallel

$m_1 = m_2$

Perpendicular

$m_1 \cdot m_2 = -1$

Line through a point

$y - y_1 = m(x - x_1)$

Angle of inclination

$m = \tan \theta$

If $\tan \theta$ is negative the calculator returns a negative angle; add 180° to get the inclination, which is measured anticlockwise from the positive x-axis and is never negative.

ƒ FUNCTIONS & GRAPHS

Function	Shape	Key features
$y = a(x + p)^2 + q$	parabola	turning point $(-p, q)$ · axis of symmetry $x = -p$ · $a > 0$ opens up
$y = \frac{a}{x + p} + q$	hyperbola	asymptotes $x = -p$ and $y = q$ · no turning point
$y = ab^{x+p} + q$	exponential	asymptote $y = q$ · $b > 1$ grows, $0 < b < 1$ decays

The sign inside the bracket is the opposite of the shift: $y = (x + 3)^2$ moves left three. Losing that costs a mark on almost every transformation question.

⊙ TRIGONOMETRY

Square identity

$\sin^2 \theta + \cos^2 \theta = 1$

Quotient identity

$\tan \theta = \frac{\sin \theta}{\cos \theta}$

Reduction

$\sin(180^\circ - \theta) = \sin \theta$

$\cos(180^\circ - \theta) = -\cos \theta$

Co-function

$\sin(90^\circ - \theta) = \cos \theta$

General solution

$\sin \theta = k \Rightarrow \theta = \sin^{-1} k + n \cdot 360^\circ$

or

$\theta = 180^\circ - \sin^{-1} k + n \cdot 360^\circ$

$\cos \theta = k$

$\theta = \pm \cos^{-1} k + n \cdot 360^\circ$

$\tan \theta = k$

$\theta = \tan^{-1} k + n \cdot 180^\circ$

The full reduction table and the CAST diagram are on the [trigonometry sheet](#). Writing $n \in \mathbb{Z}$ is worth a mark on its own.

▢ SINE, COSINE & AREA RULES

Sine rule

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

Cosine rule

$a^2 = b^2 + c^2 - 2bc \cos A$

Area rule

$\text{Area} = \frac{1}{2}ab \sin C$

Sine rule when

an angle is opposite a known side

Cosine rule when

three sides, or two sides and the angle between them

In a 3-D problem, name the triangle you are working in before writing anything down. Most lost marks here are a correct rule applied to the wrong triangle.

MEASUREMENT

Right pyramid — V	$\frac{1}{3} \times \text{base area} \times h$
Cone — volume	$\frac{1}{3} \pi r^2 h$
Cone — surface area	$\pi r^2 + \pi r \ell$
Sphere — volume	$\frac{4}{3} \pi r^3$
Sphere — surface area	$4 \pi r^2$

Multiplying every dimension by k multiplies area by k^2 and volume by k^3 . That is the whole of the "effect of change" question.

CIRCLE GEOMETRY — THE THEOREMS YOU MUST NAME

Centre to chord	A line from the centre perpendicular to a chord bisects it (and conversely)	Same segment	Angles in the same segment are equal
Angle at centre	The angle at the centre is twice the angle at the circumference on the same arc	Cyclic quadrilateral	Opposite angles are supplementary; an exterior angle equals the interior opposite angle
Semicircle	The angle in a semicircle is 90°	Tangent	A tangent is perpendicular to the radius at the point of contact; two tangents from a point are equal

Every statement in a geometry proof needs a reason, and the reason must be the theorem's name. A correct chain of statements with no reasons scores close to nothing.

R FINANCE, GROWTH & DECAY

Simple interest	$A = P(1 + in)$
Compound interest	$A = P(1 + i)^n$
Straight-line depreciation	$A = P(1 - in)$
Reducing balance	$A = P(1 - i)^n$
Nominal to effective	$1 + i_{\text{eff}} = \left(1 + \frac{i^{(m)}}{m}\right)^m$

"Depreciation on a reducing balance" is compound interest with a minus. Straight-line depreciation is simple interest with a minus. Reading which one the question means is most of the work.

Σ STATISTICS

Mean	$\bar{x} = \frac{\sum x}{n}$
Variance	$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$
Standard deviation	$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$
Interquartile range	$Q_3 - Q_1$
Outlier	below $Q_1 - 1,5(\text{IQR})$ or above $Q_3 + 1,5(\text{IQR})$
Skewed right	mean > median
Skewed left	mean < median

Use the calculator's statistics mode for σ — the formula is here so you can explain it, not so you compute it by hand under time pressure.

P PROBABILITY

Complement	$P(\text{not } A) = 1 - P(A)$
Addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Mutually exclusive	$P(A \text{ and } B) = 0$
Independent	$P(A \text{ and } B) = P(A) \times P(B)$

Independent and mutually exclusive are not the same thing and cannot both be true for events with non-zero probability. The exam sets that trap most years.

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